Research Statement

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I am an applied microeconomist working on topics in energy, environmental, and transportation economics, using tools from empirical industrial organization. My research examines how market environments, frictions, and policy incentives shape the decisions of households and firms, and how these choices affect efficiency, distributional outcomes, and environmental externalities. A central focus of my work is the clean energy transition, including the adoption of technologies that can reduce greenhouse gas emissions and local air pollution, such as electric vehicles (EVs) and solar panels (PVs), as well as the design of electricity pricing plans that influence consumer behavior. Beyond energy markets, I also investigate how endogenous rigidities, particularly contract duration, inhibit the efficient reallocation of physical capital in decentralized transportation markets, such as the global container shipping sector. Methodologically, my research combines structural estimation with causal inference techniques, informed by surveys as well as large-scale administrative data. This approach allows me to measure behavioral responses, evaluate welfare implications, and provide policy-relevant insights for regulatory design. My current research portfolio includes three papers that are under review or revision, as well as additional work in progress.

"Technology Complementarities and Subsidy Policy: Evidence from Electric Vehicle and Solar Panel Adoption"

[Job Market Paper, R&R @ Journal of the Association of Environmental and Resource Economists]

Government policies aimed at mitigating climate change and air pollution often subsidize clean energy technologies; however, they generally overlook interdependencies in technology adoption. Understanding these interdependencies can inform the design of policies that target environmental externalities more efficiently. In this paper, I focus on two important technologies, electric vehicles and residential rooftop solar panels, that both reduce emissions and may interact in significant ways. In theory, the two goods could be substitutes if adopting either one is sufficient to satisfy a household's desire to reduce its environmental footprint or to signal "green" preferences; or they could be complements if joint adoption generates additional cost savings or amplifies perceived environmental benefits. If the goods are substitutes, subsidizing one may reduce demand for the other, thereby diminishing the marginal effectiveness of subsidies. By contrast, if the goods are complements, subsidizing one can stimulate adoption of the other, amplifying emission reductions and raising the overall returns to subsidy expenditures.

I develop a novel structural multi-product demand model that incorporates interactions between electric vehicles (EVs) and solar photovoltaic (PV) systems, building on Gentzkow (2007). This is the first study to formally connect two markets through a common environmental externality, revealing how complementarities extend beyond private utility to shape social welfare. I estimate the model using household-level data from the California Vehicle Surveys. A central challenge is distinguishing true demand complementarities from simple correlation in consumer preferences. Identification comes from within-state variation in factors that affect demand for one technology but not the other, for example, a higher solar irradiation raises the value of PVs but does not directly influence the utility of EVs. I also address the classic endogeneity problem arising from correlation between prices and unobserved product characteristics using BLP-style instruments that leverage observable features of competitive market structure.

I find that solar panels and electric vehicles are complements, with the willingness to pay for joint adoption ranging from about $9,000 to $13,600, and increasing with income and vehicle size. Approximately one-third of this value reflects electricity cost savings, while the remainder is attributed to environmental preferences. Complementarity substantially raises joint adoption; without it, EV–PV bundles would be 64% less common. I show that the optimal allocation of subsidies varies with the policy target. However, when considering both global (CO_2) and local ($PM_{2.5}$) pollutants, the most efficient subsidy directs about 70% of funds toward PVs and 30% toward EVs. Thus, understanding cross-market interactions can help design policies that achieve environmental targets at lower cost to taxpayers.

“Endogenous Rigidities and Capital Misallocation: Evidence from Containerships”
(joint with Nicholas Vreugdenhil and Nahim Bin Zahur)

[R&R @ Journal of Political Economy]

In this paper, we examine how endogenous rigidities, specifically contract duration, impact the reallocation of physical capital. We focus on the global containership leasing market, a crucial part of international trade, as most goods are transported by sea and nearly half of the fleet is leased under fixed-term contracts. We construct a novel dataset of ship-level contracts and port-call records from 2005 to 2015. Using these data, we demonstrate that contract duration is pro-cyclical (in booms, firms sign longer leases) and that contract lock-in, as well as search frictions, restrict capital reallocation in downturns, leading to misallocation. We develop and estimate a new empirical dynamic matching framework with booms and busts, in which shipowners and charterers jointly choose contract length. Our model quantifies misallocation in the decentralized equilibrium at 5.6% of industry profits on average relative to the constrained social planner solution, with significant variations over the cycle. Moreover, we find that search frictions significantly amplify contracting externalities; removing them would raise welfare by 41.5%. Counterfactuals further reveal that endogenous rigidities substantially reduce the effectiveness of maritime subsidies, a common industrial policy in this sector. Overall, this study highlights the importance of accounting for endogenous rigidities when analyzing capital reallocation over the business cycle.

“Optimal Second-best Menu Design: Evidence from Residential Electricity Plans”
(joint with Nicolai Kuminoff, Spencer Perry, and Nicholas Vreugdenhil)

[Under review]

In this paper, we investigate how to design a menu of electricity price plans for residential customers to maximize social welfare in a second-best setting where the marginal private and external costs of generating electricity vary over time, institutional constraints prevent mandating time-varying pricing, and consumer behavior is distorted by frictions. Using rich administrative data covering five years of 15-minute interval electricity consumption for more than 8,000 households, along with information on demographics and rate plan enrollment, we document four descriptive facts about consumer behavior. First, by implementing a quasi-experimental design that leverages longitudinal and cross-sectional variation in prices, we show that a subset of consumers responds strongly to time-of-use (TOU) plan incentives by load shifting and load shaving. However, our second finding is that most consumers do not respond to marginal prices. Third, we document selection into plans on both the level and slope of demand. Lastly, switching plans is hindered by inattention, as consumers tend to change plans only after receiving unusually high bills. Building on these facts, we develop a two-stage structural model: in the first stage, households choose a rate plan, and in the second stage, they decide how much to consume in every 15-minute interval of a day and how to shift the timing of their consumption during the day in response to perceived price incentives. Using our model, we find that social welfare would increase if existing TOU plans were removed from the choice set, since the large temporal shifts in consumption they trigger generate little social benefit but impose substantial adjustment costs on households. Next, we show that a hypothetical information treatment that increases the rate at which consumers respond to marginal prices would further decrease welfare by exacerbating inefficient load-shifting responses to current plan incentives. Finally, we solve for the second-best optimal menu of price plans. We demonstrate that a simple menu consisting of the utility's existing block rate plan and a newly redesigned TOU plan can achieve 80% of the benefits of the theoretical ‘first-best’ menu.

Ongoing and Future Research

I am starting a new project with Nicholas Vreugdenhil and Nahim Bin Zahur that extends our prior work on the container shipping market to examine “**Environmental Externalities from Contract Rigidities and Search Frictions**.” The shipping industry accounts for about three percent of global CO_2 emissions and is a significant contributor to local air pollution, including $PM_{2.5}$ and NO_x, particularly in port regions. Despite improvements in fuel efficiency, total emissions from maritime transport have been rising, driven by the expansion of global trade and inefficiencies in fleet utilization. The framework naturally links economic inefficiency to emissions: in booms, when ships are highly utilized, the carbon intensity of trade (CO_2 per ton-mile) declines; in downturns, however, long-term contracts and search frictions create lock-in and underutilization, increasing emissions per ton-mile. Moreover, when firms remain locked into long-term contracts, they may be unable to reallocate to newer, cleaner vessels as fuel prices or environmental regulations evolve, suggesting that contract-induced misallocation can slow the diffusion of low-emission technologies. It would be interesting to estimate how contractual and search frictions increase "empty miles at sea" and to quantify how much of the cyclical variation in maritime emissions can be attributed to these frictions rather than to changes in trade volume. The framework would enable an analysis of the effectiveness of various policy instruments aimed at reducing emissions in this industry. For example, carbon taxes or fuel subsidies may interact with contracting behavior: by internalizing fuel costs, these policies could encourage firms to sign shorter or more flexible contracts, enhancing market responsiveness and potentially lowering emissions. Furthermore, policies that promote contract flexibility or facilitate early vessel reallocation, such as green leasing standards or retrofit subsidies, could accelerate the adoption of cleaner technologies. Finally, improving coordination and matching mechanisms between ships and charterers could further mitigate excess emissions by reducing ship underutilization.

I am also continuing to work on the design of electricity rates. I expect to complete a new working paper in the near future on "**Optimizing Electricity Rates: Evidence from Commercial and Industrial Firms**." In this study, I focus on the commercial and industrial (C&I) sector, which accounts for more than half of electricity demand yet has received limited attention in prior research. Using administrative data on 5,000 C&I customers' 15-minute electricity consumption and rate plan history, I document: (i) most firms could reduce their electricity expenditures by switching rates; (ii) higher-consumption firms are more likely to select their cost-minimizing rate, while low-consumption customers could save a higher percentage of their bill by switching; (iii) most new customers avoid TOU plans despite the potential cost savings; and (iv) a subset of firms that switch to TOU pricing reduce peak-period demand, with significant heterogeneity across industries. Building on these findings, I extend my prior work on modeling residential consumption to model rate selection and hourly consumption in the C&I sector, incorporating operational flexibility, price responsiveness, and adjustment frictions. This framework captures technological and organizational constraints that limit how quickly firms can adjust electricity use compared to households. I am currently estimating the model and will use it to assess the welfare implications of current C&I price plans, as well as counterfactual menus under alternative scenarios for future expansion of the manufacturing and services sectors (e.g., more data centers).

In the longer run, I would like to extend my **future research** to link my demand-side work with the supply side to study the electricity market as an integrated system. On the supply side, I am interested in studying endogenous generation and emissions costs, capacity investment decisions, and the implications of shifts in the generation mix driven by the rapid expansion of renewable energy and large-scale storage. As renewable generation expands, the variance of hourly energy output increases, leading to greater price volatility and short-term uncertainty. Greater demand flexibility, enabled by smart interval meters, automated technologies, and dynamic pricing, changes how short-term fluctuations affect long-term reliability. In the short term, flexible consumers can smooth demand and reduce the likelihood of supply shortages; in the long term, this flexibility lowers the need for generation capacity and influences investment decisions. Understanding these dynamics has the potential to improve the design of electricity markets that maintain reliability and support decarbonization at the lowest possible cost.

Ideally, I would like to develop a framework to study the optimal joint design of retail electricity rates and generation investment decisions. Such a framework could be used to quantify how behavioral flexibility can substitute for physical generation or storage capacity while maintaining reliability standards, and to analyze how changes in demand affect investment incentives, equilibrium capacity mix, and long-run welfare. Moreover, by modeling both supply and demand, one could compare the efficiency of subsidizing production versus consumption. For example, is it more effective to allocate subsidies to utility-scale renewable generation, distributed technologies such as rooftop solar and battery storage, or demand-side incentives for shifting the timing of consumption?

Finally, I would like to examine the distributional consequences of large load entry under various rate structures, identifying the conditions under which incumbent consumers, particularly low-income or low-usage customers, bear higher costs. I think it could be interesting to evaluate the equity and efficiency of mechanisms such as differential pricing for large versus small customers, entry-based infrastructure charges, TOU pricing, and demand charges.